Supply Chain Management & Integrated Planning

This course is based on the definitions and planning concepts formalized by APICS. It also attempts to integrate certain practices and performance indicators used within the Lean framework.

It is useful to consult in parallel with this course:

- *Chapters 1, 2, 3, and 5 of J. R. Tony Arnold, Stephen N. Chapman, and Lloyd M. Clive, Introduction to Materials Management (Boston et al.: Pearson, 2012)*
- *For vocabulary details: APICS Dictionary, 2013, 14th edition*

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1. CORPORATE STRATEGY AND SUPPLY CHAIN MANAGEMENT

1.1 Concept of Value

In general, Supply > Demand ⇒ Focus on the Added Value expected by the Customer

In design :
Value concept in the sense of value analysis:

$$Value = \frac{satisfaction\ of\ needs}{ressources\ consumed}$$

In production/logistics management:

$$Value\ for\ the\ Customer = \frac{Quality \times Service}{Cost \times Lead\ time}$$

- Quality: features, performance level, and technical specifications of an offer
- Service: availability, support, and commitments provided to the customer
- Cost: total cost for the customer: price + all associated costs throughout the product life cycle (usage, maintenance, storage, management, technical support, training, recycling/disposal, etc.)
- Lead Time: time required to meet customer requirements (i.e., delivery lead time).

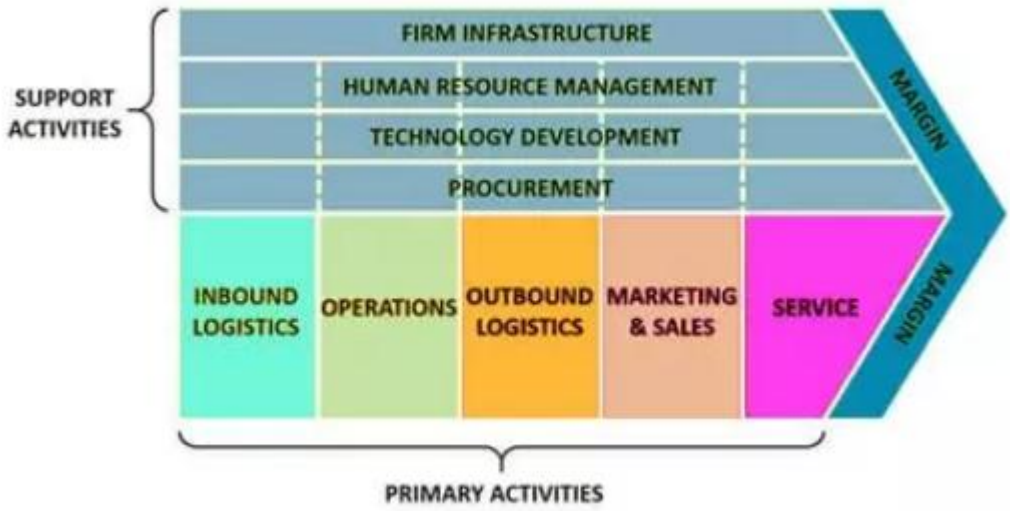
In Lean terms :

- **Value Added: what the customer is willing to pay for** — any activity that transforms materials, services, or information to directly meet customer expectations

- Non–Value Added: what the customer is not willing to pay for — any activity that does not increase the value of the product or service and that adds cost, time, resource usage, or space

How is value created within the organization?

Figure 1 : Porter's Value Chain



Primary Activities

- Inbound logistics: upstream logistics activities such as receiving, storing, and internal handling
- Operations: transformation of materials and components into finished products
- Outbound logistics: activities related to delivering goods and services to the customer
- Marketing and sales: methods and means used to make the company's offering known, appreciated, and purchased
- Service: activities associated with the main offering (training, maintenance, etc.)

Support Activities

- Procurement: activities related to purchasing materials, goods, supplies, and production resources
- Technology development: includes information systems, R&D, knowledge management, etc.
- Human resource management: all activities related to recruitment, compensation, motivation, training, and career management
- Firm infrastructure: general management and other functions commonly called “support,” such as accounting, legal, etc.

Competitive advantages?

		Competitive Advantage	
		Lower Costs	**Differentiation**
Competitive Scope	**Broad target** / Entire industry	1. **Cost leadership**	2. **Differentiation leadership**
	Narrow target / Particular segment	3A. **Cost-focus**	3B. **Differentiation-focus**

Figure 2 : Porter’s Four Generic Strategies

Cost leadership strategy**:** : offering a product or service with the same perceived value as competitors, but at a lower price.

Two types of **Differentiation Leadership** strategies**:**

- **Upward differentiation (sophistication)**: offering a more sophisticated product than the reference offer, and selling it at a higher price. Ideally, the price increase should exceed the cost increase in order to generate higher profit.
- **Downward differentiation (stripping down)**: offering a less elaborate product than the reference offer, but selling it at a lower price. Ideally, the cost reduction should exceed the price reduction in order to generate higher profit.

Downward differentiation	**≠**	**Cost leadership.**
Reduced value and reduced price		Reduced price (but maintained value)

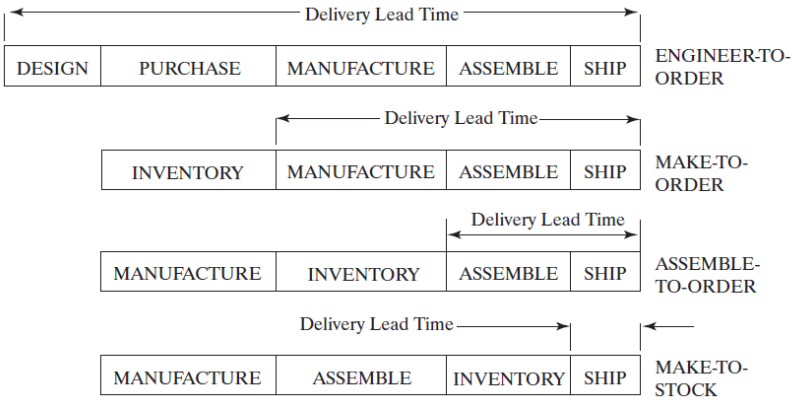
Focus strategy = “niche strategy”: concentrating most efforts on a small market segment in order to avoid direct confrontation with stronger competitors → within this niche, choosing either a cost-leadership strategy or a differentiation strategy.

Order winners[1] ≠ Order qualifiers[2]

[1] **Order winners**
Competitive characteristics of a company that encourage customers to choose its products or services rather than those of its competitors.
Order Winners focus on one (rarely more than two) of the following strategies: price (cost), quality, delivery time, delivery reliability, product design, flexibility, after-sales service, and brand image.

[2] **Order qualifiers**
Competitive characteristics a company must possess to be a viable competitor in the market.

1.2 Different Supply Chain Management Strategies



Strategy choice? Delivery time / customer lead time (**Delivery / Global Lead Time**[3]) ?
Customer order penetration point[4] / decoupling point ?[5]

Make To Stock (MTS)	Make To Order
/ Product demand characteristics	
Relatively stable and predictable demand Very short customer lead time required by the market Low-cost storable product (shelf-manageable) Few options	Product/service made according to customer specifications (≠ market-specific) Customer can wait (for how long?) High production and storage cost Product with options
/ Production & procurement process characteristics	
➔ Target service level? ➔ Production system flexibility? In volume? In variety? ➔ Product configuration & delayed differentiation? Type of bill of materials? ➔ Parts of the process with significant value-added contribution? ➔ Outsourced processes? ➔ Diversity of components / production steps & lead times for replenishment?	
Input data	
Demand forecast over the planning horizon? Starting inventory level? Ending inventory level? Back orders?	Demand forecast over the planning horizon? Initial backlog (orders on hand) level? Ending backlog (orders on hand) level? Back orders?

[3] **Delivery lead Time**
Time elapsing between the acceptance of a customer order and its delivery (or shipment), and possibly until payment is received.

[4] **Order penetration point**
A key element of a logistics organization, it is the point (moment) at which a product is assigned to a specific customer. Downstream from this point, the system is driven by customer orders; upstream, processes are driven by forecasts and planning programs.
This point is different from the decoupling point, since the customer order may be placed earlier or later.

[5] **Decoupling point**
In a production process, the last stage to which a product—usually launched based on forecasts in the MPS—can be advanced and stored (this is the decoupling stock) while waiting for an order (final manufacturing order or firm customer order), typically coming from the Final Assembly Schedule (FAS).
This stage enables the start of the final transformation steps of the product through final assembly, customization, packaging, or any other operation required to complete the product before making it available to the customer.

2. PLANNING SYSTEM

2.1 Overall Structure of the Planning System

What do we need to produce? (Demand? What, how much, when?) What is required to make it? (Workload / required capacity?) What resources do we have? (Available capacity?) What do we need? (Adjustments?)	Purpose of the plan? Planning horizon? Level of detail and aggregation? Update frequency?

Different levels of a production planning system?
At each level, workload vs. resource capacity?

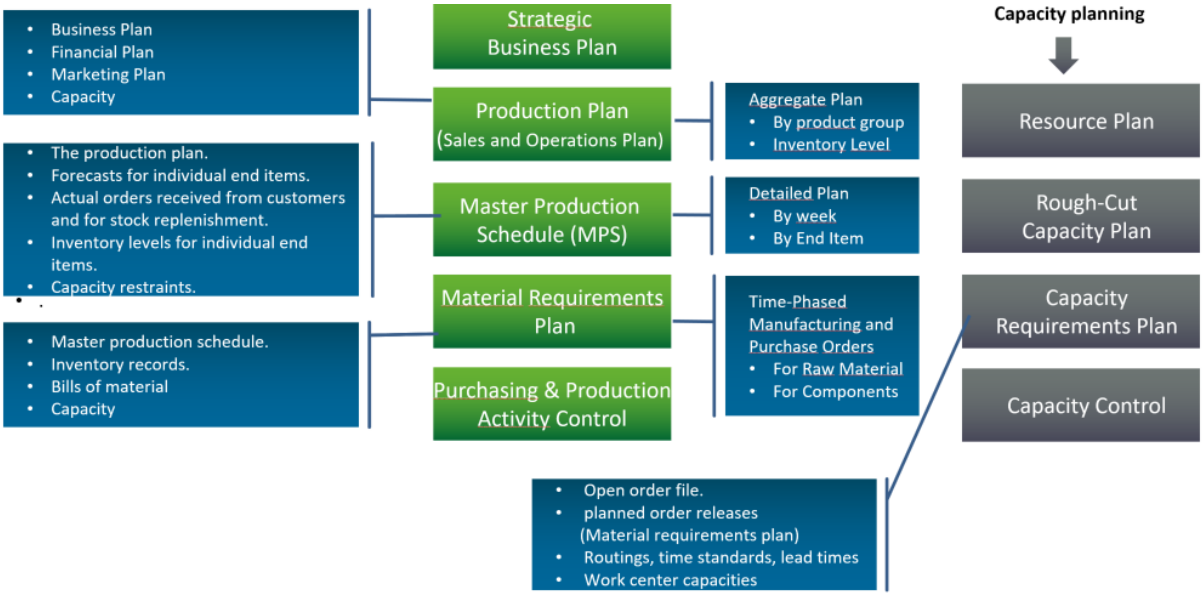


Figure 3 : Production Planning and Control System

Strategic business plan (Plan stratégique)
A plan that defines the main goals and objectives an organization intends to achieve in the coming years, as well as the major strategic directions to reach them (which future markets, with which product ranges, produced in which plants and with which partners, etc.).

Sales and Operations Planning – S&OP (Planification Industrielle et Commerciale - PIC) :
Based on the company's Strategic Plan (profit, productivity, customer service level, market share), **this level of planning defines and sets the overall conditions for providing the product/service offering, within a given budget and resource framework.**

MPS - Master Production Schedule (PDP - Programme Directeur de Production) :
It takes into account forecasts, the Sales and Operations Plan, the order portfolio, material and resource availability, and management objectives, in order to establish the reference framework for Production over a given period.

Material Requirement Planning (MRP) (Planification des besoins en matière et composants)
A set of techniques that use bills of materials, inventory status, and the Master Production Schedule to calculate material and component requirements.

Net Requirements Calculation (Calcul des Besoins Nets)
Processes and techniques that make it possible to convert gross requirements into net requirements, using bills of materials, inventory status, and the MPS (Master Production Schedule) to calculate material or component needs.

Production Activity Control – PAC (Pilotage d'atelier, Suivi de production) :
Function to manage and schedule the work to be performed at the level of a production unit and to manage suppliers. This encompasses all the principles, approaches, and techniques necessary to schedule, monitor, measure, and evaluate the effectiveness of production activities.

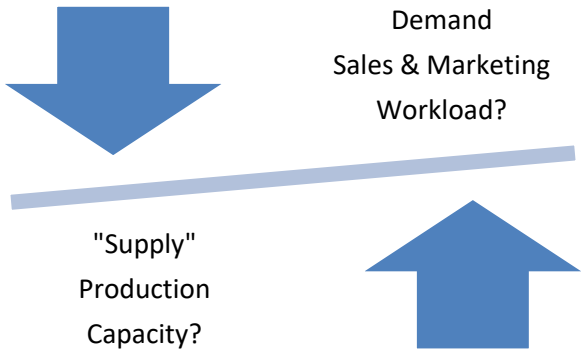
Dispatching (Ordonnancement / lancement)
Selection, sequencing, and assignment of operators to tasks to be performed at individual workstations.

2.2 Production Plan – Sales and Operations Plan (S&OP)

The strategic plan is generally updated annually → The S&OP (Sales and Operations Planning) process allows for regular:

- Updating data from the strategic plan to coordinate all activities/services/departments within an organization.
- Anticipating upcoming changes with a sufficiently frequent update cycle.
- Providing a realistic plan to achieve overall objectives (costs, service levels, productivity, etc.).
- Better managing production levels, inventory, and order backlog.[6]

➔ **Level of aggregation?** Product family/range / a limited number of products
➔ **Choice of a common unit?** (number of products (output), hours, tons, €, etc.)
➔ **Horizon?** Long enough to implement planned changes (HR, equipment, long lead-time replenishments, etc.) & shorter than the lead time for major capacity changes (new product, plant, strategic partner, major investments) – i.e., 6 to 18 months
➔ **Update frequency? ↗** depending on market turbulence (seasonality?), demand uncertainty (known horizon?), and responsiveness of production and supply systems – i.e., monthly, sometimes weekly



Demand[7] ≠ Sales ; Expected Demand [8] ≠ Actual Demand [9] ; Dependent Demand[10] ≠ Independent Demand[11]

[6] **Backlog (Carnet de commande)**
All customer orders received but not yet delivered.

[7] **Demand (demande)**
Requirement for a specific component or product.
The sources of demand can be diverse: customer orders, sales forecasts, inter-plant requests, replenishment needs for a warehouse (spare parts), or requirements for the production of another product.
For a finished product, demand may differ from sales because demand does not necessarily translate into sales (e.g., if there is no stock to meet it, no sale will occur). Demand generally has four components: trend, cycle, seasonality, and random variation.

[8] **Expected Demand (Demande attendue)**
The quantity of products expected to be withdrawn from stock during the total lead time if product consumption follows the forecast (equal to the forecast for the period).

[9] **Actual Demand (Demande réelle)**
These are customer orders (and item reservations for production and distribution) that "consume" the forecasts, according to chosen rules, over a given horizon.

[10] **Dependent Demand**
This refers to the level (quantity/value) of demand for an item that depends on the availability/provision of other items; such demand is calculated and should not be forecasted.

[11] **Independent Demand**

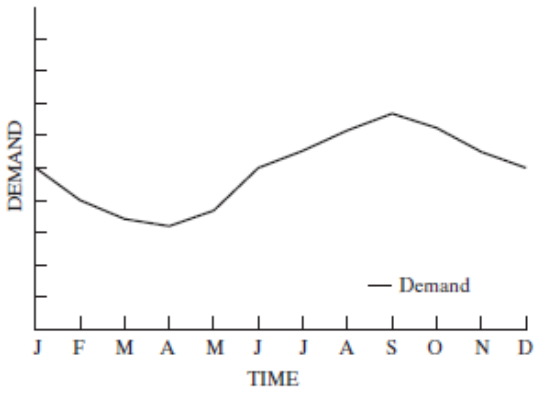
Demand Management (for products) → Workload? → Required Capacity? (= capacity demand) → Available Capacity? → Capacity Management

2.3 Different Planning Strategies

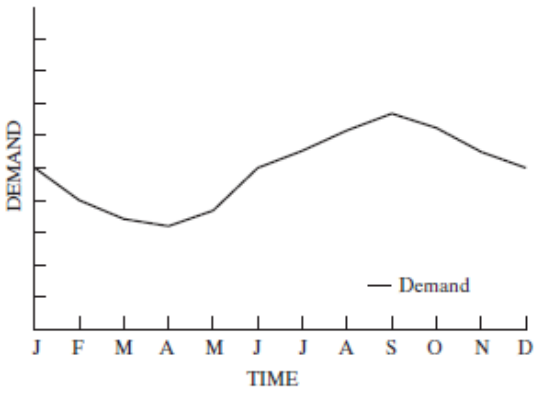
Possible levers at the S&OP level: :

- Hire/fire staff, increase or decrease the number of shifts, overtime, or idle periods (underload).
- Increase or decrease inventory levels (coverage/turnover[12]) or order backlog.
- Increase or decrease subcontracting, number of leased equipment

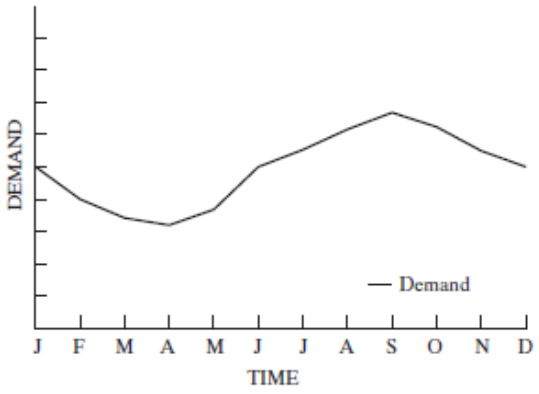
➔ Generally, three basic strategies are considered:



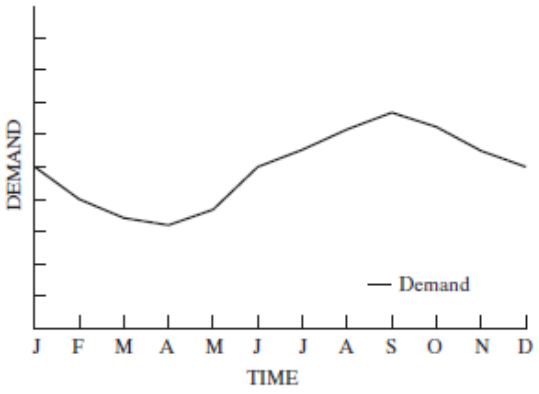
chase strategy ; demand matching strategy
(stratégie de production à la demande)



level strategy
(stratégie de nivellement/lissage de la prod.)



subcontracting strategy
(stratégie de sous-traitance)



hybrid strategy
(stratégie hybride)

Production Smoothing in Make-to-Stock (MTS)

1. Determine the value of the cumulative forecasted demand over the planning horizon.
2. Establish the starting **inventory** level and set the target ending inventory at the end of the planning horizon.
3. Calculate the total production to be planned over the planning horizon:
   Cumulative Production = Cumulative Forecasted Demand + Backorders + Ending Inventory – Starting Inventory
4. Calculate the planned production for each period:
   Planned Production per Period = Cumulative Production / Number of Periods in the Planning Horizon
5. Calculate the ending inventory for each period:
   Ending Inventory = Starting Inventory **+** *Planned Production per Period* **–** *Planned Demand*

In this case, the demand for the item is not linked to any other acquisition, production, or distribution event of other products.

[12] **Stock coverage**: the period during which inventory can meet forecasted demand
Inventory turnover: Consumption during a period T / Average inventory during period T

Production Smoothing in Make-to-Order (MTO)

1. Determine the value of cumulative forecasted demand over the planning horizon.
2. Establish the number of **backlog orders** at the start and set the target backlog at the end of the planning horizon.
3. Calculate the total production to be planned over the planning period:
   Cumulative Production = Cumulative Forecasted Demand + Backorders + Starting Backlog – Ending Backlog
4. Calculate the planned production for each period:
   Planned Production per Period = Cumulative Production / Number of Periods in the Planning Horizon
5. Calculate the planned backlog at the end of each period:
   Planned Backlog at Period End = Starting Backlog **–** *Planned Production for the Period* **+** *Planned Demand for the Period*

➔ **Concept of Takt Time**

Takt time is the ideal pace to meet customer demand.
Takt time allows for the establishment of leveled production schedules over one or two weeks for easily reconfigurable production systems, and up to one year for less flexible systems.

$$\boldsymbol{Takt\ Time} = \frac{\boldsymbol{Operating\ Time\ of\ the\ Workshop\ or\ Production\ Lin}}{\boldsymbol{Customer\ Demand}}$$

Change in Production Level → Change in Takt Time → Reconfiguration of the Production System

$$\boldsymbol{Number\ of\ Workstations\ /\ Stations\ /\ Operators} = \frac{\boldsymbol{Work\ Content}}{\boldsymbol{Takt\ Time}}$$

Note: The work content corresponds to the workload assigned to a given resource.

➔ **OPR**

OPR : Overall Production Ratio
L'OPR measures the ability to achieve the planned quantities within the available time:

$$\boldsymbol{OPR} = \frac{\boldsymbol{Number\ of\ good\ pieces} \times \boldsymbol{Takt\ Time}\ (\boldsymbol{or\ Target\ Cycle\ Time})}{\boldsymbol{Total\ available\ time}}$$

The OPR is measured based on the line cycle time (not the standard cycle time of a machine, operator, or workstation).

OPR allows evaluating the efficiency of a production line and is therefore similar to OEE (Overall Equipment Effectiveness), which measures the efficiency of a machine.

It is not a productivity indicator, since it does not include machine hours or direct labor hours in the denominator, but rather time or total hours.

3. CAPACITY MANAGEMENT

3.1 Concept of Capacity

Capacity is the ability to produce a good or a service (i.e., having the necessary resources to meet demand).

Workload ≠ Capacity of a Resource

- **Workload of a production resource**: the number of production hours or items corresponding to a demand that is decided to be produced on a given production resource.
- **Capacity of a production resource**: the number of production hours or items that can potentially be produced within a given period, under certain working assumptions.

→Both concepts are similar to a rate or throughput.

capacity required (capacité requise) ≠ capacity available (capacité disponible)

≠ budgeted capacity (capacité budgétée)

demonstrated capacity (capacité démontrée) ≠ rated capacity (capacité calculée/nominale /effective)

→ Required capacity is similar to a workload.

3.1.1 Utilization and Efficiency

Set based on the analysis of past production data and/or target improvements for these two indicators over time (usually at the Sales & Operations Planning – S&OP – level).

The **Utilization** measures the intensity (usually as a percentage) with which a resource is used for production.

$$Utilisation = \frac{Hours\ Actually\ Worked}{Available\ Hours}$$

Efficiency measures adherence to predefined standards:

$$Efficiency = \frac{Standard\ hours\ produced}{Hours\ actually\ worked}$$ [13]

Note :

- Spending more time indicates lower efficiency.
- Efficiency can be greater than 100% → indicates a higher demonstrated capacity → standard times should be reviewed.
- In contrast, productivity measures production relative to the resources used.
- Efficiency measures adherence to the schedule when calculated based on planned hours rather than actual hours worked.

3.1.2 Available Capacity

Rated Capacity corresponds to the production volume that a production system or resources can handle over a given period.

$$\boldsymbol{Rated\ capacity\ (Capacité\ calculée) = Available\ time \times Utilization \times Efficiency}$$

Lead time ≠ Workload (note: they can be equal if only a single resource is considered)

[13] **hours actually worked (Heures réellement travaillées)**

Available capacity evolves based on:

- Product specifications → changes result in modifications to the work content.
- Product mix → changes result in modifications to the overall work content.
- Equipment used.
- Production pace/effort; e.g., if the Takt Time changes, line balancing changes, and more broadly, line efficiency (OPR) may be affected.

Demonstrated capacity: obtained through the analysis of past production data.

3.1.3 Required Capacity

Required Capacity (i.e., workload) corresponds to the total number of hours needed (expressed in standard hours) to complete all work orders (WOs) in progress and planned within a given scope (machine, workstation, line, plant) over a defined period.

Calculation of Required Capacity:

1. Determine the operation times in standard hours for all WOs currently in production or planned, for each resource considered in each planning period.

$$Total\ Standard\ Time\ =\ Setup\ time\ +\ Run\ time$$

2. Sum all these standard hours for each resource considered over each planning period.

Next, the "actual" required hours are determined, which correspond to the estimated number of hours truly needed to meet demand, taking into account efficiency losses and anticipated underutilization of resources.

$$\boldsymbol{Actual\ time}\ (\boldsymbol{Heures}\ \text{réellement}\ \boldsymbol{requises}) = \frac{\boldsymbol{Capacity\ required}\ (\boldsymbol{in\ Standard\ hours})}{\boldsymbol{Utilization} \times \boldsymbol{Efficiency}}$$

3.2 Different Levels of Capacity Planning

Related to the different levels of production and sales planning (cf. Figure 3: Production Planning and Control System)

Resource planning (Planification des ressources)
Capacity planning conducted at the strategic level (can be based on the strategic plan and/or the S&OP). Activities involve establishing, measuring, and adjusting long-term capacity limits and levels. This concerns resources whose acquisition requires a long lead time. Decisions made at this level require top management approval.

Rough Cut Capacity Planning (Vérification des charges globales) :
Procedure that involves translating the Production Plan and/or Master Production Schedule into capacity requirements for critical resources: labor, machines, floor space, financial resources, etc.

CRP - Capacity Requirement Planning (Planification des besoins en capacité) :
Function of establishing, measuring, and adjusting capacity limits or levels, particularly determining the amount of resources (labor and machines) required for production tasks.
The released and planned orders from the requirements calculation serve as inputs to the CRP (Capacity Requirements Planning), which converts these orders into working hours per workstation and per period.

Capacity Control (Gestion des capacités) :
Process that involves measuring actual production and comparing it to the planned capacity requirements, checking whether deviations exceed predefined limits, and taking corrective actions.

3.3 OPT (Optimized Production Technology)- Théorie des Contraintes

1. Balance the Flow, Not the Capacities

It is impossible, even by adjusting them, to balance a company's capacities. Instead, the flow must be regulated to achieve smooth throughput. For this, the entire flow must be aligned with that of the bottleneck.
Note: In processes that are well-balanced but have high variability in operations or demand (especially in terms of product mix), the bottleneck can constantly change. This sometimes leads to creating a known and controlled "forced" bottleneck; the Takt Time is then adjusted according to the bottleneck.

2. A Time Loss at a Bottleneck is a Loss for the Whole System

Since the bottleneck has no spare capacity, it cannot recover lost time. Conversely, providing additional capacity at the bottleneck benefits the entire system.

3. Time Savings on a Non-Bottleneck is Illusory

Non-bottleneck resources remain dependent on the bottleneck. An hour gained on a non-bottleneck does not benefit the system; it only increases work-in-progress or inventory.

4. The Utilization of a Non-Bottleneck is Determined by System Constraints, Not Its Own Potential

By applying the Drum-Buffer-Rope (DBR) principle, non-bottleneck resources must be synchronized with the bottleneck. Using non-bottlenecks at full capacity would only produce excessive WIP or inventory.
The Five Key Steps to Implement DBR:

- Identify the constraint (the bottleneck).
- Exploit the constraint (maximize its utilization and efficiency).
- Subordinate all other processes to the constraint.
- Elevate the constraint's performance (increase its capacity → hiring, investment, etc.).
- Return to Step 1 if the constraint has changed.

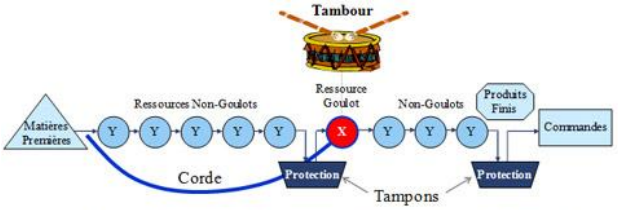


Schéma : Le pilotage des flux selon la technique "Tambour – Tampon – Corde" de la Théorie des Contraintes

5. Resource Utilization and Activation Are Not Synonyms

Do not manage a resource based on local productivity indicators, but based on the production flow.
Activation: using a resource to produce (aiming for full employment of the resource).
Utilization: using a resource to produce according to the production flow.

6. Bottlenecks Determine Both Output and Inventory Levels

The overall system throughput cannot exceed that of the constraint. Blindly activating upstream resources can quickly inflate inventory.

7. Transfer Batch and Production Batch Do Not Have to Be Equal

A transfer batch smaller than the production batch allows parallel production across multiple resources and enables faster handoff to the next resource, accelerating the flow.
One-piece flow: transfer flow piece by piece.

8. Production Batches Should Be Variable, Not Fixed

Instead of defining so-called "economic" batch sizes, which can harm flexibility and proper bottleneck loading, the Theory of Constraints recommends dynamically determining batch size based on demand. Batch size is calculated at each launch rather than being fixed.
Some processes are not flexible, and their setup times cannot be neglected; in such cases, batch sizes may remain large (see SMED). A compromise must then be made between throughput losses on one hand and on-time delivery and WIP increase on the other.

9. Establish Schedules Considering All Constraints Simultaneously

Planning and scheduling should be exclusively based on the bottleneck. It must always be working. Ensure that upstream capacities and material availability are consistent.

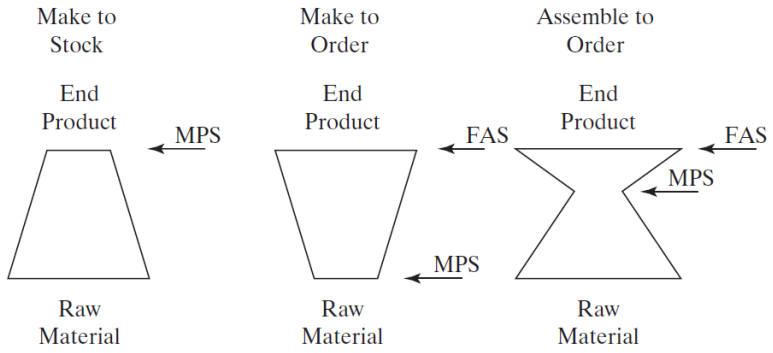
TOC Motto:
The sum of local optima does not equal the global optimum.

4. MASTER PRODUCTION SCHEDULE (MPS) - PLAN DIRECTEUR DE PRODUCTION (PDP)

4.1 Development of the Master Production Schedule (MPS)

Structuring the MPS

The MPS should be developed at the stage where production diversity and options are at their lowest:



Objective? To drive the requirements planning process (see MRP / Material Requirements Planning)
Level of detail and aggregation? Which master schedule item(s) (articles directeurs) should be used?
Planning horizon? ≥ the longest cumulative lead time[14]
In ATO (Assemble-to-Order) environments,
the FAS (Final Assembly Schedule) horizon [15] ≥ the assembly lead time of the ordered product (the time required to manufacture components is not included).
Update frequency? How often the MPS should be reviewed and refreshed.

Evaluation of the MPS based on three criteria:

- Resource use
- Customer Service compliance
- Costs

Information required to develop the MPS:

- Production Plan (from the S&OP)
- Forecasts for each finished product (over a shorter horizon than those used in S&OP)
- Firm customer orders or orders related to stock replenishment
- Inventory levels for each finished product / planning item
- Updated capacity constraints

Steps for Preparing the MPS:

1. Develop the preliminary MPS
   The forecasted demand for each period comes from the Production Plan (i.e., after disaggregating the "Planned Production – Production Plan" line) and incorporates updated forecasts and the current order book for each planning item.
2. Compare the preliminary MPS with available capacity
   Are the critical resources (production bottleneck, critical labor or equipment, scarce resources, or long lead-time resources) available?
3. Resolve issues related to gaps between the preliminary MPS and available capacity
   Possible actions include: hiring, overtime, rerouting to less critical resources, subcontracting, etc.

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[14] **Délai cumulé (Critical path Lead Time, cumulative/aggregated lead time)**
Planned longest lead time for a given activity. It is obtained by examining the lead time of each branch of the item's multi-level bill of materials. The longest path represents the cumulative lead time.

[15] **Final Assembly Schedule (FAS) : Programme de Montage/d'Assemblage Final (PMF ou PAF)**
Program that determines the finalization of the product based on customer orders in a make-to-order or assemble-to-order environment. It is also called the Finishing Schedule because it may include operations other than final assembly. It may also involve no assembly at all, but only mixing, cutting, or packaging operations. It is prepared after receiving a customer order and validated after analyzing material and equipment availability. The FAS schedules the finishing operations of the product from the level at which its components are stocked (or planned in the MPS) up to the finished product..

	On Hand (Stock init)	Periods 1	2	3	4	5
Forcast - Demande prévisionnelle		5	5	8	10	15
Projected available Balance - Stock Dispo	20					
MPS - PDP						

Lot Size: 30

Safety Stock: 5

Tableau 1 : MPS without ATP

Notes on the terminology used:
On-hand inventory (Stock physique) :
At a given moment, the quantity of items physically present in the actual stock or recorded as present in the inventory files (including items already sold but awaiting shipment).

Available inventory (Stock disponible)
On-hand inventory minus allocations, reservations, backorders, and (usually) quantities blocked for quality issues.

Reservation (Réservation)
Items assigned to the execution of a specific order or schedule but not yet withdrawn from inventory.
It is a “non-debited” requisition.
Allocation (Affectation)
Procedure used to distribute an item when availability is limited.

Projected available inventory ou Projected available balance ((Stock) disponible prévisionnel)
Inventory projected into the future. For a given period, it is equal to:
Projected available=Projected on-hand + Expected receipts + Planned orders − Gross requirements

In the MPS grid:

Projected available = Projected on-hand + MPS − Sales forecast

Note:
At the MPS level, safety stock corresponds to additional planned inventory and capacity intended to protect against forecast errors and short-term fluctuations in the order book.

4.2 Taking Available-to-Promise (ATP) into account

ATP (Available-to-Promise) is the quantity actually available for new customer orders. The term can be translated as “Disponible à la Vente” (DAV) in French.

Steps to complete the MPS grid with ATP calculation:

1. Calculate the projected available inventory for each period in the planning horizon:
   Projected available inventory = Prior period Projected available inventory + MPS* − Max (forecast, customers orders)
2. Calculate Available-to-Promise (ATP)
   - Using discrete logic:
     - For the first period:
       ATP = Initial on-hand inventory + MPS − Sum of customer orders scheduled before the next MPS**
     - For each subsequent period where a production order is expected to be completed:
       ATP = MPS − Sum of customer orders scheduled before the next MPS**

- Selon une logique cumulative
  - For the first period:
    ATP = Initial on-hand inventory + MPS − Sum of customer orders scheduled before the next MPS**
  - For each subsequent period where a production order is expected to be completed:
    ATP = Previous ATP + MPS* − Sum of customer orders scheduled before the next MPS*

** Note: This refers to the completion of planned production orders in the MPS, not the start of production orders.*

		Periods				
	On Hand (Stock init)	1	2	3	4	5
Forcast - Demande prévisionnelle		5	5	8	10	15
Order - Commandes Client		**5**	**3**	**2**	**0**	**0**
Projected available Balance - Stock Dispo	20					
Available to Promise						
MPS - PDP				30		

Lot Size: 30

Safety Stock: 5

Tableau 2 : MPS with ATP established using a discrete logic

		Periods				
	On Hand (Stock init)	1	2	3	4	5
Forcast - Demande prévisionnelle		5	5	8	10	15
Order - Commandes Client		**5**	**3**	**2**	**0**	**0**
Projected available Balance - Stock Dispo	20					
Available to Promise						
MPS - PDP				30		

Lot Size: 30

Safety Stock: 5

Tableau 3 : MPS with ATP established using a cumulative logic

5. SUPPLY CHAIN MANAGEMENT

Supply chain management - Gestion de la chaine Logistique
Design, planning, execution, control, and monitoring of supply chain activities to create value, develop a competitive distribution network, implement international logistics, synchronize supply and demand, and measure overall performance.

Note: The supply chain covers the entire product lifecycle.

5.1 Structuring and Modeling the Supply Chain

Supply chain - Chaine logistique
Global network for delivering products and services from raw materials to end customers through **integrated flows of information, physical goods, and finances**.

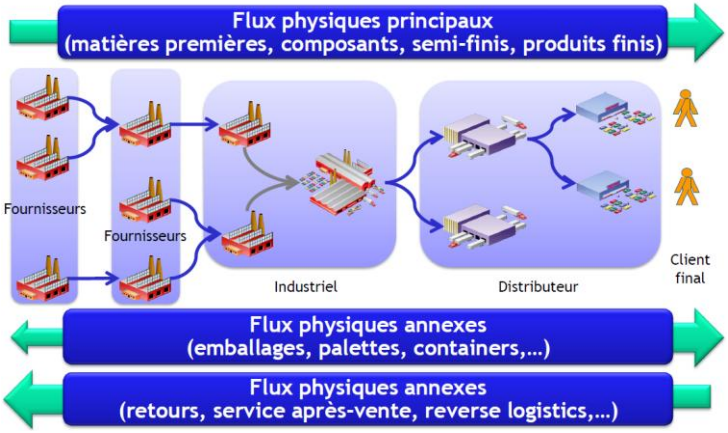


Figure 4 : Flux Physiques

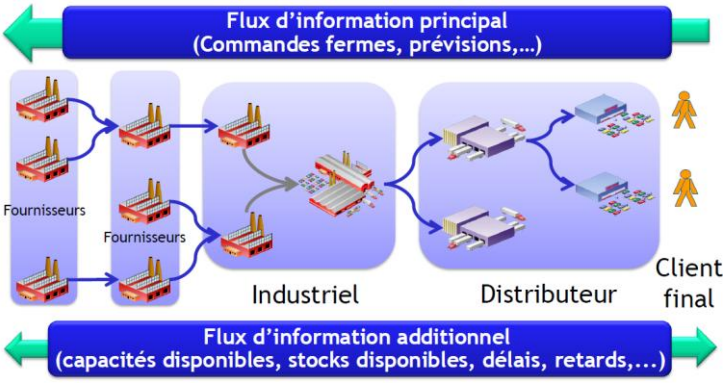


Figure 5 : Flux d'Information

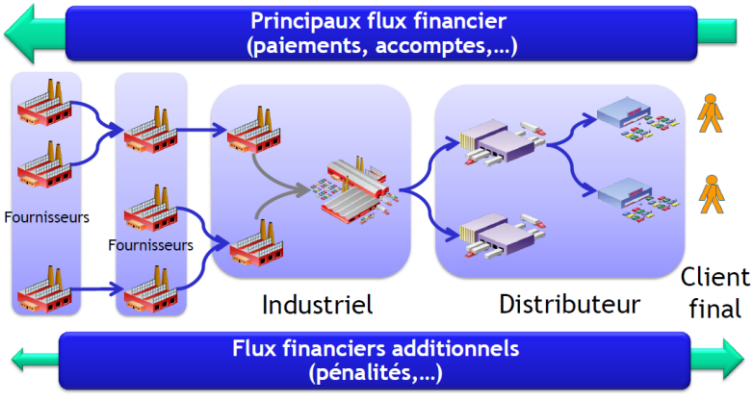


Figure 6 : Flux Financiers

Logistics system - Système Logistique
Planning and coordination of a company's physical movements, such as the flows of raw materials, components, and finished products, to minimize costs while achieving the desired service level.

Bullwhip effect - Effet coup de Fouet
Amplification of demand variability upstream in the supply chain, caused by a minor change occurring downstream.
Stocks can quickly fluctuate from shortage to surplus.
This phenomenon is due to the cascading effect of orders moving up the chain and product lead times moving down the chain.

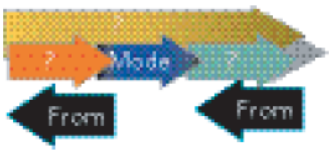
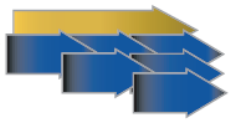
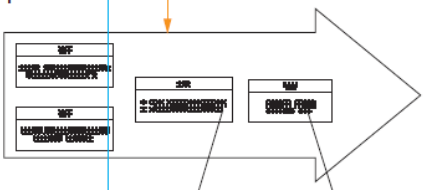
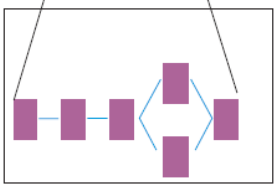
The bullwhip effect can be eliminated by synchronizing the supply chain.

Supply-Chain Operations Reference Model (SCOR)

Developed by the Supply Chain Council (SCC), the SCOR model (Supply Chain Operations Reference Model) is a standardized methodology for describing and evaluating processes within a supply chain.

The SCOR model describes supply chain operations across four levels:

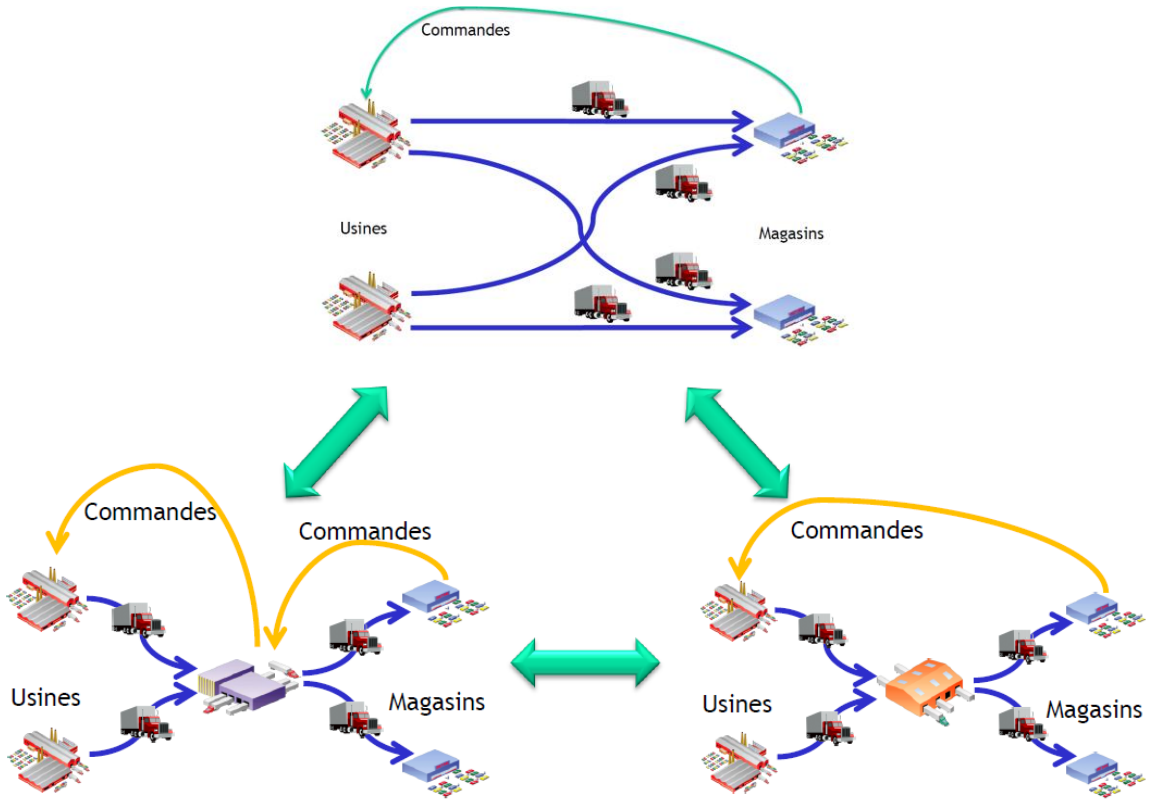
- Level 1: Identifies five generic processes: Plan, Source, Make, Deliver, and Return.
- Level 2: Defines categories within each process reflecting the company's chosen operational strategy, such as direct or indirect distribution, make-to-order or make-to-stock production.
- Level 3: Details the steps of each process category identified at Level 2, including their inputs and outputs.
- Level 4: Describes company-specific procedures.

	Niveau	Description	Schématisation	Commentaires
Supply Chain Operations References Model	1 	Top niveau		• Cadre et contenu de la Supply Chain • Opérations Référence Model • Ici sont définies les bases des objectifs de performance à atteindre
	2 	Niveau de configuration		• La SC peut être configurée « sur commande » au niveau 2 à partir de 30 processus principaux • Les entreprises mettent en place leur stratégie opérationnelle à travers la configuration de leur SC
	3 	Niveau de décomposition		• Le niveau 3 définit la capacité de l'entreprise à être compétitive sur son marché • Les entreprises affinent la stratégie des opérations à ce niveau
Not in Scope	4 	Niveau de mise en place		• Les entreprises mettent en place des pratiques managériales spécifiques à ce niveau • Le niveau 4 définit les pratiques pour atteindre un avantage concurrentiel et s'adapter aux conditions changeantes

The model evaluates four dimensions:

- Reliability of business performance
- Flexibility / Responsiveness
- Supply chain cost
- Return on invested capital

Direct Distribution (Factories → Stores) vs Distribution via Storage Warehouses vs Distribution via Cross-Docking Platforms



Cross-docking - Transbordement (de quai à quai)
Concept of splitting and allotting incoming product shipments to facilitate sorting at intermediate warehouses or delivery according to final destinations.
Items are transferred from the receiving dock to the shipping dock without intermediate storage in the warehouse.
Cross-docking helps reduce inventory levels and storage capacity requirements.

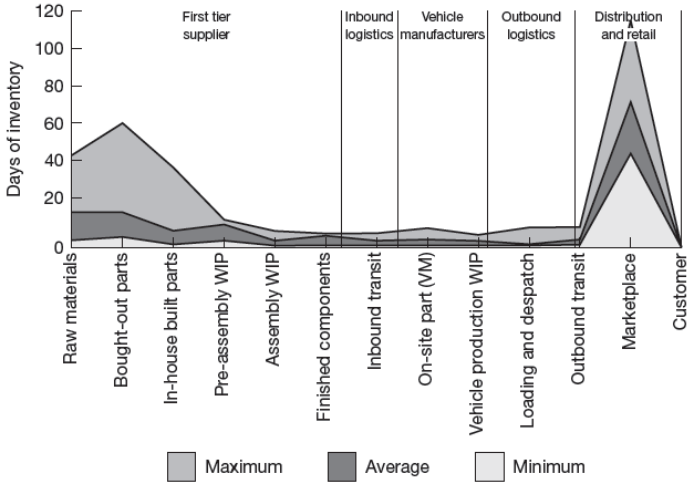


Figure 7 : Inventory Coverage Profile in the Automotive Industry[16]

[16] Source: Holweg, M. and Pil, F.K., The Second Century, MIT Press, 2004

5.2 Distribution Requirement Planning (DRP)

Notes on the terminology used:

Distribution Requirement Planning (DRP I)[17] ≠ Distribution Ressource Planning (DRP II)[18]

Forcast (Prévision)[19] ≠ Gross requirements (Besoins bruts)[20]
Scheduled receipt (Réception attendue)[21] ≠ Planned order receipt (Réception planifiée)[22]
Planned order release (Lancement d'un ordre planifié)[23]
≠ Firm planned order(FPO) (Ordre planifié ferme)[24]

Rappel : Net requirements (Besoins nets)
$Net\ Requirements = Gross\ Requirements + Reservations - Expected\ Receipts - Available\ Inventory$
$= Gross\ Requirements + Reservations - Expected\ Receipts - (On - hand\ Inventory - Safety\ Stock)$
After lotting and time-phasing, net requirements become planned orders.

MRP	DRP
Scheduled receipt (Réception attendue) Procurement / manufacturing Lead Time (Délais d'approvisionnement / de fabrication)	≈ in-transit inventory (Stock en transit) [25] In-transit lead time (Temps de transit)[26]
Projected available inventory ou Projected available balance ((Stock) disponible prévisionnel) Planned order receipt (Réception planifiée) Firm planned order(FPO) (Ordre planifié ferme) Same for MRP & DRP	

Figure 8 : Analogies between MRP and DRP grids

[17] **Distribution Requirement Planning (DRP I) – Calcul des besoins nets de distribution**
–

[18] **Distribution Ressource Planning (DRP II) – Planification des Ressources de Production**
Extension of net requirements calculation for distribution to the planning of critical resources in the distribution system: warehouse capacity, labor, costs, trucks, etc.

[19] **Forcast (Prévisions)**
Estimation of future demand. It can be obtained using quantitative methods, qualitative methods, or a combination of both.
Forecasting can be based on exogenous (external) or endogenous (internal) factors.
Different forecasting methods aim to predict one or more components of demand: trend, cycle, seasonality, and random variations.

[20] **Gross requirements (Besoins bruts)**
Total of dependent and independent demand for the same item before calculating net requirements, taking into account available inventory and expected receipts.

[21] **Scheduled receipt (Réception attendue)**
Released (open) order, to which a due date is assigned.

[22] **Planned order receipt (Réception planifiée)**
Quantity expected to be received as a result of a planned order release.
Planned order receipts differ from expected receipts because planned orders have not yet been released.

[23] **Planned order release (Lancement d'un ordre planifié)**
Row in a net requirements calculation table that is derived from planned receipts by offsetting the corresponding quantity by the appropriate lead time.

[24] **Firm planned order(FPO) (Ordre planifié ferme**
A planned order whose quantity and timing are "frozen".
It cannot be modified automatically by the computer system, but only by the planner responsible for the item. This technique allows MRP users to address inventory or capacity issues by "freezing" selected planned orders.
Firm orders are normally the way to express the Master Production Schedule (MPS).

[25] **In-transit inventory (Stock/marchandise en transit)**
Product moving between two or more locations, usually geographically separated (e.g., finished goods shipped from a factory to a distribution center).

[26] **In-transit lead time (Temps de transit)**
The time elapsed between the shipment date and the receipt date at the dock.
Since orders usually specify the date on which goods must be delivered to the dock, subtracting this transit time allows determination of the supplier's shipment date.

Warehouse A		1	2	3	4	5	6	7	8
Forecasts		130	120	110	150	160	130	140	110
In transit			250						
Projected available inventory	285								
Planned receipts									
Planned shipments									

Reorder point	100
Lead time	2
Order quantity	250

Warehouse B		1	2	3	4	5	6	7	8
Forecasts		80	90	110	100	80	90	60	70
In transit		150							
Projected available inventory	110								
Planned receipts									
Planned shipments									

Reorder point	50
Lead time	1
Order quantity	150

Logistics platform		1	2	3	4	5	6	7	8
Forecasts		0	400	150	250	400	0	150	0
In transit			300	300					
Projected available inventory	300								
Planned receipts									
Planned shipments									

Reorder point	125
Lead time	3
Order quantity	300

Tableau 4 : Example of a DRP Grid

5.3 Collaborative Approaches

Vendor-Managed Inventory (VMI) - Gestion Partagée des Approvisionnements (GPA)
A method for optimizing supply chain performance in which the supplier has visibility of the customer's inventory and is responsible for maintaining its levels. The supplier replenishes stock based on regular analysis of the customer's inventory data.
Customer inventory is calculated, taking into account damaged or expired products, and replenished to predefined levels. The supplier receives a receipt for the products delivered to replenish inventory and invoices the customer accordingly.

CPFR (Collaborative Planning, Forecasting and Replenishment) - Gestion collaborative de la planification, de la prévision et des réapprovisionnements

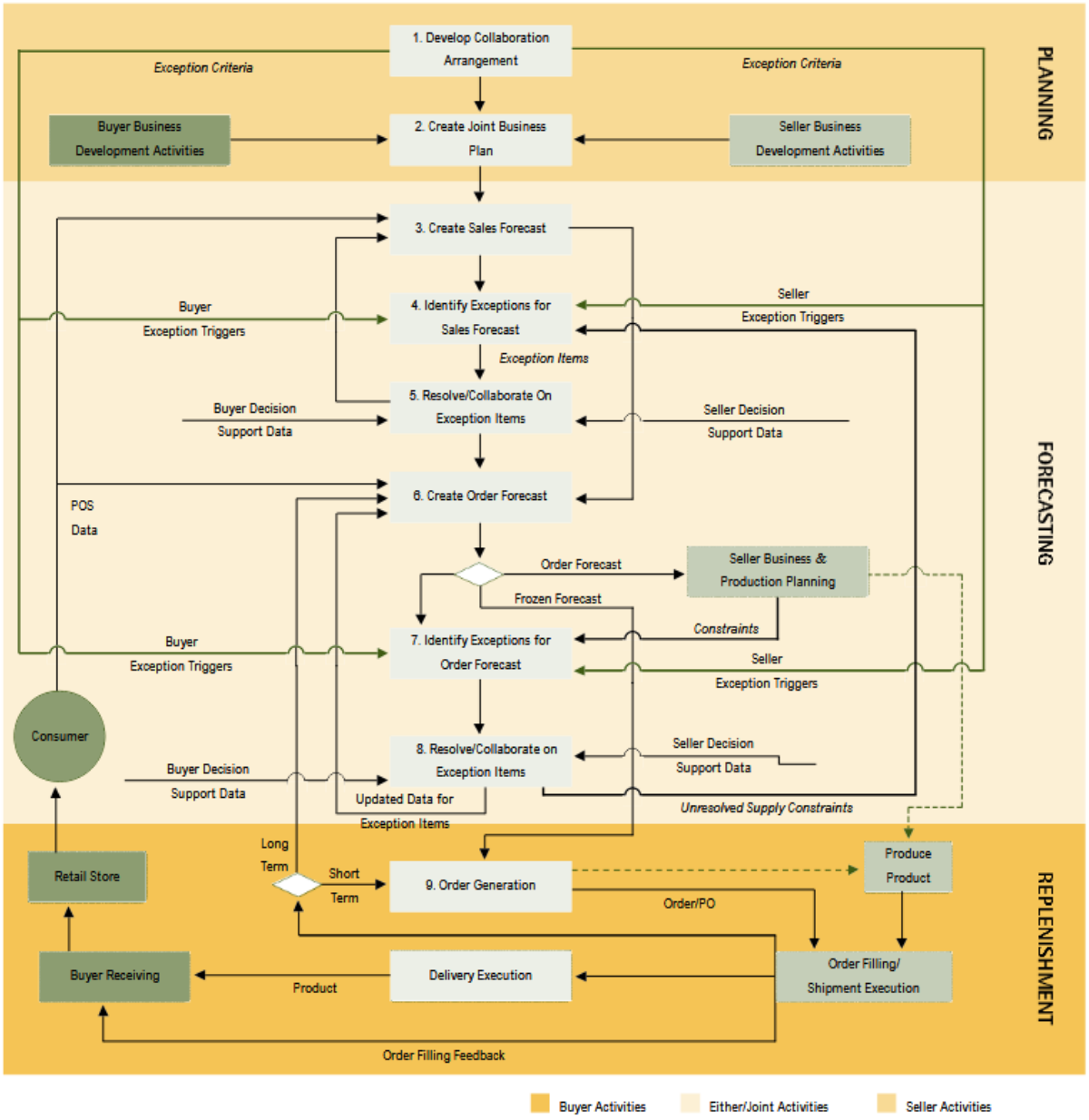


Figure 9 : The 9 Steps of the CPFR Model